

Approach to EEG Reading

Let's start. I recommend this stepwise approach. The first step is the same as with any test, confirm the *patient's identity*. Be sure the EEG you are about to read belongs to the correct patient. Additionally, note three important points before reading. The technologist will usually mark these out for the reader.

1. *Age*: The EEG is relatively stable during adulthood. However, certain normal variants are common in young adults. A slight slowing of the posterior-dominant rhythm may be considered within normal after the age of 80, but it must remain within alpha range.
2. *State(s)*: The EEG varies with state changes; it is important to note if the patient was awake, drowsy, or asleep during the record. Furthermore, epileptic discharges are accentuated during drowsiness and sleep. If the patient is confused or comatose, then the EEG is reflective of that state. More on this later.
3. *Skull defect*: The skull serves as a natural filter of higher frequencies. EEG activity over a skull defect such as a craniotomy is typically sharper, faster, and of higher amplitude compared to the rest of the skull. This is called a breach effect. *To avoid overcall, the sharpness of a waveform must be interpreted with caution over a skull defect* [1]. Most epileptic discharges are negative (upward) but with the alteration of neuronal dipoles from brain surgery, such as a temporal lobectomy, the discharges may be positive (downward) [2].

Next, open to view the record.

Before interpreting the EEG activity, *check the settings*. These are usually placed on the top in most digital displays. They include the *recording parameters* (*filters, sensitivity, paper speed, or time base*), the *montage*, and the *calibration signal*. If you have read the first part of the book, you are already well versed with each of these terms.

Now, let's get to the exciting part: the brain waves!

Spend a few seconds staring at the first page. Observe and admire the myriad of squiggly lines that makes up the EEG. You're looking at your patient's cerebral cortex in action and there is no other test that can do this. As you look at the page, some activities will become obvious, they will catch your eye. This is what we shall refer to as the foreground. In contrast, the remainder of the page is the background. You must do this each time you see a record, until it becomes second nature. Remember, the foreground is what stares back at you when you look at the page and the rest is all background. Some beginners may find this exercise difficult at first: I did. Our mind can be easily overwhelmed by the sheer amount of activity on a page, which makes separating foreground from the background confusing.

I can assure you, that these difficulties will resolve with time and practice. One helpful exercise is to apply this concept to artwork. Go look at a few paintings. Ask yourself "Who are the main protagonists in the scene?," that's foreground, the scenery behind them is background. One of my all-time favorite paintings is in Figure 9.1, *The Fog Warning* by Winslow Homer, painted in 1884. It now resides in the Museum of Fine Arts in Boston [3].

What do you see? The fisherman, his boat, and catch are the foreground. The sea, the waves, their crests, the approaching fog, and the sky are the background. Try this yourself on some of your favorite paintings. In no time you will be a master at distinguishing the foreground from the background.

Now, let's come back to the EEG.



Figure 9.1 Winslow Homer, *The Fog Warning/Halibut Fishing*, 1885. Oil on canvas.

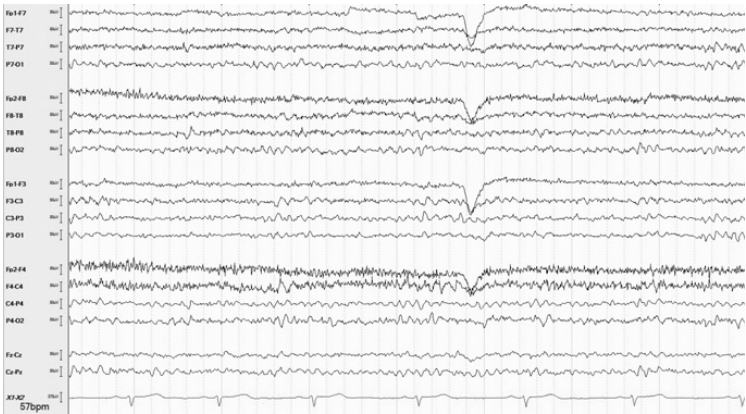


Figure 9.2 Thirty-two-year-old gentleman with an intact cranium.

In Figure 9.2 is the EEG of a 32-year-old man in normal wakefulness with an undamaged cranium. During your moment of studied reflection, mentally note the large positive (downward) deflections in the frontal channels (Fp1-F7, Fp1-F3) and (Fp2-F8, Fp2-F4), also note the short fast spiky pattern that is seen in channels Fp2-F4 and F4-C4 (Common electrode F4). These are the foreground features on this page and the remainder is background. In Figure 9.3, I have marked out the components of the foreground in this page. Currently, there is no need to worry about what these waveforms mean. All you need to be able to do, is *tell the foreground, apart from the underlying background*.

Once you have identified the foreground and the background, we will describe each of these in further detail. Conventionally, most readers will first describe the background. Arguably, it is the more important part of the record, as it tells us the state of the underlying cortex. Therefore, let's first concentrate on the background.

To describe the background, use six key features [4]. They are as follows:

1. symmetry
2. continuity
3. voltage
4. organization
5. reactivity or variability
6. sleep architecture

We will learn about each of these in Chapter 10.

Once we have described the background, we come to the *foreground*.

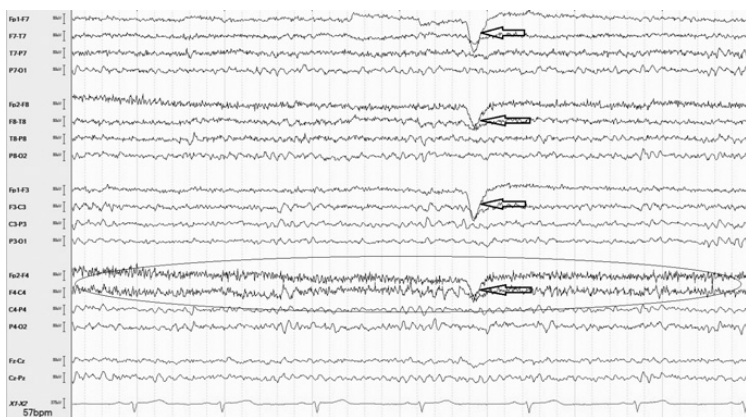


Figure 9.3 Foreground components include the large positive deflections (arrows) in the frontopolar electrodes (Fp1 and Fp2) and the short fast spiky pattern (circled) in channel F4.

Foreground components should be categorized as either cerebral activity or artifact.

1. *Activity*: Activity refers to cerebral activity. These waveforms are generated by the cerebral cortex. Cerebral activity should be described by *location* (generalized, lateralized, focal, or multifocal), *occurrence* (sporadic or repetitive), and *morphology* (slow waves or sharp/spike waves). You will learn to identify and describe each of these waveforms in Chapter 11.
2. *Artifacts*: Artifacts are contaminants. These patterns are not produced by the brain. External or nonphysiological artifacts are those that originate from outside sources, such as the ringing of a telephone. Internal or physiological artifacts originate from within the patient's body, but from sites other than the brain, such as eye blink artifact or EKG artifact. The main purpose of recognizing artifacts is to avoid confusing them with cerebral activity. Chapter 12 describes commonly encountered artifacts.

After describing the cerebral activity, assign a motive, is it normal or abnormal?

1. *Normal variants*: These are normal patterns that tend to stand out and appear abnormal. Many occur during drowsiness and sleep. Recognizing these variants avoids mistaking them for abnormalities. Chapter 13 describes normal variants.

2. *Abnormalities*: Cerebral activity that indicates an underlying abnormality should be described as mentioned, that is, by location, occurrence, and morphology, that is, slow or sharp waves.
 - i. Certain abnormalities are associated with epileptic seizures/epilepsy; these are called epileptogenic abnormalities. Epileptogenic abnormalities are usually sharp in appearance (spike or sharp wave), hence they may also be called epileptiform abnormalities (or epileptic discharges). However, this isn't absolute. Certain slow rhythmic patterns, which are not epileptiform, may also be epileptogenic abnormalities. For example, temporal intermittent rhythmic delta (TIRDA) and occipital intermittent rhythmic delta activity (ORIDA) are associated with epileptic seizures. Sporadic abnormalities are described in Chapter 14, while repetitive abnormalities are described in Chapter 15.
 - ii. So, what is a seizure? When an abnormal pattern transforms its speed (frequency), shape (morphology), or spreads (location) it is called *evolution*. Evolution is the electrographic hallmark of a seizure, but all evolution isn't a seizure. Note that I purposely left out amplitude; changes in amplitude alone do not constitute evolution in the context of whether a pattern could be a seizure. Evolution must be in either frequency, morphology, or location. Waveforms that represent ongoing seizure activity are called ictal patterns (Chapter 16).

Finally, we must know the *activation procedures*. These are performed to provoke epileptic activity. The two most commonly performed procedures are *photic stimulation* and *hyperventilation* as described in Chapter 17.

Just before you finish, there are a few final features to look at:

1. *EKG channel*: This is usually a single channel electrocardiogram that is useful to detect abnormalities in heart rhythm.
2. *Technologist's notes*: Always make a point to simultaneously look through the technologist's log notes as you interpret the EEG. These include information such as a change in the patient's mental state, movements, or other pertinent clinical descriptions. Looking at the *video*, if available, also allows the reader to make clinical correlations [5].

Using the above approach may seem a bit tedious at first, but with a little practice, the entire exercise will become intuitive.

You will be able to rapidly analyze the EEG.

Chapter Summary

Go through the following checklist when reading an EEG:

1. Confirm the patient's identity, age, state(s) of recording, and the presence of any skull defects.
2. Confirm the technical parameters of including the filter settings, sensitivity, paper speed, or time base. Note the montage you are reading in and the calibration signal.
3. Identify the background from the foreground.
4. Describe the background based on symmetry, continuity, voltage, organization, reactivity (or variability), and sleep architecture.
5. Categorize the foreground components as cerebral activity or artifact.
6. Describe cerebral activity based on its location (general, lateral, or focal), occurrence (sporadic or repetitive), and morphology (slow or sharp).
7. Then categorize the activity as normal (normal variant) or abnormality.
8. Decide whether the abnormality is epileptogenic (associated with seizures) or not. If epileptogenic, is it ictal (ongoing seizure). Epileptogenic abnormalities are not always epileptiform (spike or sharp waves). Evolution is the hallmark of electrographic seizure activity. Remember that isolated changes in amplitude are not evolution.
9. Look for the use of any provocation methods, such as hyperventilation and photic stimulation, and their effect on the EEG.
10. Before you finish up, make sure you've looked at the single channel EKG, technologist's log, and video (if available).

Now you can correlate your findings with your patient's clinical presentation.

References

1. Cobb WA, Guiloff RJ, Cast J. Breach rhythm: the EEG related to skull defects. *Electroencephalography and Clinical Neurophysiology*. 1979 Sep 1;**47**(3):251–71.
2. Otsubo H, Steinlin M, Hwang PA, et al. Positive epileptiform discharges in children with neuronal migration disorders. *Pediatric Neurology*. 1997 Jan 1;**16**(1):23–31.
3. Museum of Fine Arts, Boston.
4. Hirsch LJ, LaRoche SM, Gaspard N, et al. American clinical neurophysiology society's standardized critical care EEG terminology: 2012 version. *Journal of Clinical Neurophysiology*. 2013;**30**(1):1–27.
5. American Electroencephalographic Society Ad Hoc Guidelines Committee. Minimum technical requirements for performing clinical electroencephalography. *Journal of Clinical Neurophysiology*. 1994;**11**:2–5.